

MX17 DREAM DAQ Returned to CEA Saclay

Oct 5, 2026 · Dylan Neff & Stephan Aune

The MX17 DREAM DAQ was returned to CEA on 5 October 2026 and tested the same day: every FEU powered up, communicated with the TCM and produced normal pedestals.

Background

The DAQ was assembled in 2026 for the MX17 Micromegas tracker at CERN n_TOF, where it read out the X17 search in July and August 2026. It was returned on 5 October 2026 by Jean-Yves Rousse, together with everything else borrowed for the campaign, including all DREAM cables and other small items.

Test

1. **Power-on.** Everything was plugged in and switched on; all units turned on.
2. **One FEU at a time.** The low-voltage supply used at n_TOF was rented from CERN and went back there, so a smaller 10 A LV unit was used. It can power the TCM and only one FEU at once, so each of the 9 FEUs was tested in turn with the TCM.
3. **Pedestals, no load.** With no detector cables plugged in, a 1000-event pedestal run was taken on each FEU (13:27–13:40). This checks that each FEU communicates as expected once powered, and that every DREAM ASIC gives a reasonable pedestal.

Results

All 9 FEUs and all 4608 channels (72 DREAM ASICs) gave normal pedestals.

- **Communication:** each FEU was read out on its own, and the FEU ID in its data stream matched the ID the configuration assigns to it.
- **Data integrity:** 9,546 events in total, none missing and no lost samples.
- **Pedestals:** baselines between 194 and 453 ADC on every channel.
- **Noise:** median 3.0 ADC raw and 2.4 ADC after common-noise subtraction, equal across FEUs to within 3 %. These low values are expected with no detector attached.

MX17 DREAM DAQ back at CEA — pedestals on all 9 FEUs, 5 October 2026

4608 channels, 9,546 pedestal events, one FEU at a time · no channel more than 2× or less than ½× its FEU's median noise



Figure 1. Every channel of the 9 FEUs side by side. Top: pedestal baseline. Bottom: noise before (orange) and after (blue) common-noise subtraction. The repeating pattern is the 64-channel structure of each DREAM ASIC, identical on every card.

Conclusion

Everything looks normal. The DAQ was returned to CEA Saclay on 5 October 2026 and verified to be in working condition on the same day.

Compiled with Claude (Anthropic)